

EE3005 Communication Systems - Assignment 2

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Topics covered: Amplitude modulation, frequency/phase/angle modulation, and demodulation techniques.

Instructions: Submit your solutions in one PDF file with all step-by-step answers, plots and outputs written/printed in that file.

1. Use a small-error PLL analysis to show that a first order loop ($H(s) = 1$) cannot track an incoming signal whose instantaneous frequency is varying linearly with time ($\theta_i = kt^2$). This signal can be tracked within a constant phase if $H(s) = (s + a)/s$. It can be tracked with a zero phase error if $H(s) = (s^2 + as + b)/s^2$.
2. Consider a superheterodyne FM receiver designed to receive the frequency band of 1 to 30 MHz with an IF frequency 8 MHz. What is the range of frequencies generated by the local oscillator for this receiver? An incoming signal with a carrier frequency of 10 MHz is received at the 10 MHz setting. At this setting of the receiver, we also get interference from a signal with some other carrier frequency if the receiver RF-stage bandpass filter has poor selectivity. What is the carrier frequency of the interfering signal?
3. An angle-modulated signal with carrier frequency $\omega_c = 2\pi \times 10^5$ is described by the equation

$$\varphi(t) = 5 \cos(\omega_c t + 20 \sin(1000\pi t) + 10 \sin(2000\pi t))$$

what is

- a) Peak phase deviation
 - b) Instantaneous frequency deviation.
 - c) Bandwidth
4. You are asked to design a DSB-SC modulator to generate a modulated signal $km(t) \cos(2\pi f_c t + \theta)$, where $m(t)$ is a signal band-limited to B Hz. The figure shows a DSB-SC modulator available in the stock room. The carrier generator available generates not $\cos(2\pi f_c t)$, but $\cos^3(2\pi f_c t)$, Explain whether you would be able to generate the desired signal using only this equipment. You may use any kind of filter you like.

- What kind of filter is required ?
- Determine the signal spectra at points b and c , and indicate the frequency bands occupied by these spectra.
- What is the minimum usable value of f_c ?
- Would this scheme work if the carrier generator output were $\sin^3(2\pi f_c t)$? Explain.
- Would this scheme work if the carrier generator output were $\cos^n(2\pi f_c t)$ for any integer $n \geq 2$?

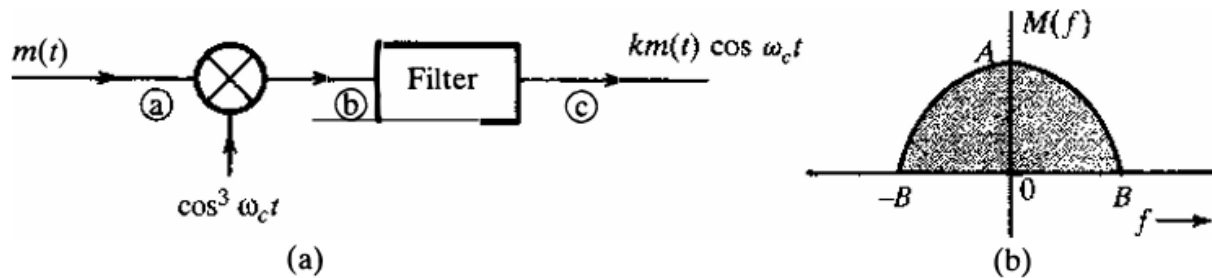


Figure 1: DSB-SC modulator

- The input $m(t)$ to an FM modulator with $k_f = 1$ has Fourier transform

$$M(f) = \begin{cases} j2\pi f, & |f| < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

The output of the FM modulator is given by

$$u(t) = A \cos(2\pi f_c t + \phi(t)),$$

where f_c is the carrier frequency.

- Find an explicit time-domain expression for $\phi(t)$ and carefully sketch $\phi(t)$ as a function of time.
 - Find the magnitude of the instantaneous frequency deviation from the carrier at time $t = \frac{1}{4}$.
 - Using the result from part (b) as an approximation for the maximum frequency deviation, estimate the bandwidth of $u(t)$.
- A voltage-controlled oscillator (VCO) has a quiescent frequency of $f_c = 1$ GHz and a frequency sweep of 2 MHz/mV. It produces an angle-modulated signal whose phase deviation $\theta(t)$ from the carrier is shown in Figure below.

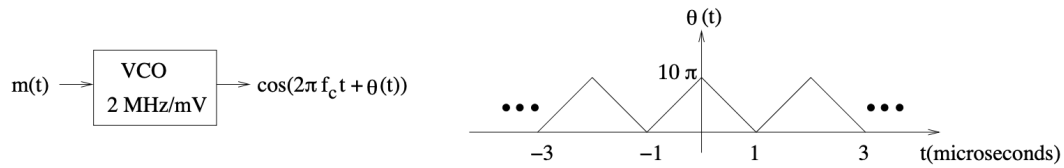


Figure 2: Phase deviation $\theta(t)$ of the VCO output signal.

- (a) Sketch the input $m(t)$ to the VCO, carefully labeling both the voltage and time axes.
- (b) Estimate the bandwidth of the angle-modulated signal at the VCO output. You may approximate the bandwidth of a periodic signal by that of its first harmonic.

7. The modulating signal that is the input to an FM modulator is

$$m(t) = 10 \cos(16\pi t).$$

The output of the FM modulator is

$$u(t) = 10 \cos \left[4000\pi t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau \right],$$

where $k_f = 10$. The output of the FM modulator is passed through an ideal bandpass filter (BPF) centered at $f_c = 2000$ Hz with a bandwidth of 64 Hz.

Determine the power of the frequency components at the output of the filter. What percentage of the transmitter power appears at the output of the BPF?

8. Consider

$$u(t) = 20 \cos(2000\pi t + \phi(t)),$$

an angle-modulated signal where

$$\phi(t) = 0.1 \cos(2\pi t).$$

Let

$$y(t) = u^{12}(t).$$

- (a) For $\phi(t) = 0.1 \cos(2\pi t)$, estimate the approximate bandwidth of $u(t)$.
- (b) Specify the output when $y(t)$ is passed through an ideal bandpass filter (BPF) centered at 12 kHz.
- (c) Using Carson's formula, determine the bandwidth of the BPF required to recover most of the information in $\phi(t)$ from the output.

9. Consider a message signal

$$m(t) = \cos(2\pi f_m t + \phi)$$

and a corresponding DSB-SC signal

$$u_p(t) = A m(t) \cos(2\pi f_c t)$$

where $f_c > f_m$.

- a) Write the frequency-domain expression for lower sideband (LSB) and upper sideband (USB) signals.
- b) Sketch the spectra of the corresponding LSB and USB signals (Sketch the real and imaginary parts separately).
- c) Find the time-domain expression for LSB and USB signals.

10. An AM signal is generated by modulating a carrier

$$c(t) = A_c \cos(2\pi f_c t)$$

with a message signal

$$m(t) = A_m \cos(2\pi f_m t)$$

where $A_c = 10$, $A_m = 3$, $f_c = 1\text{MHz}$ and $f_m = 10\text{kHz}$.

- a) Write the mathematical expression of the AM signal and determine the modulation index.
- b) Determine the total transmitted power, carrier power and power in each sideband.
- c) If the modulation index is increased to 1.2, what will happen to the AM waveform and the demodulated output.

11. A single-tone FM signal is given by

$$s(t) = 10 \sin[2\pi f_c t + 20 \sin(2000\pi t)]$$

where $f_c = 8 \times 10^6 \text{ Hz}$.

- a) Determine modulation index
- b) Determine modulation frequency
- c) Determine frequency deviation
- d) Determine power of the FM signal

12. A carrier

$$c(t) = A_c \cos(2\pi f_c t)$$

is frequency modulated with a message signal

$$m(t) = A_m \cos(2\pi f_m t)$$

where $A_c = 8$, $A_m = 4$, $f_c = 1\text{MHz}$ and $f_m = 5\text{kHz}$. The frequency sensitivity of FM modulator is $k_f = 10\text{kHz/V}$.

- a) Derive the expression for the FM signal and calculate the frequency deviation and modulation index.
- b) Estimate the bandwidth of the FM signal.
- c) If the amplitude of the message signal is doubles, explain how the FM spectrum and bandwidth change.

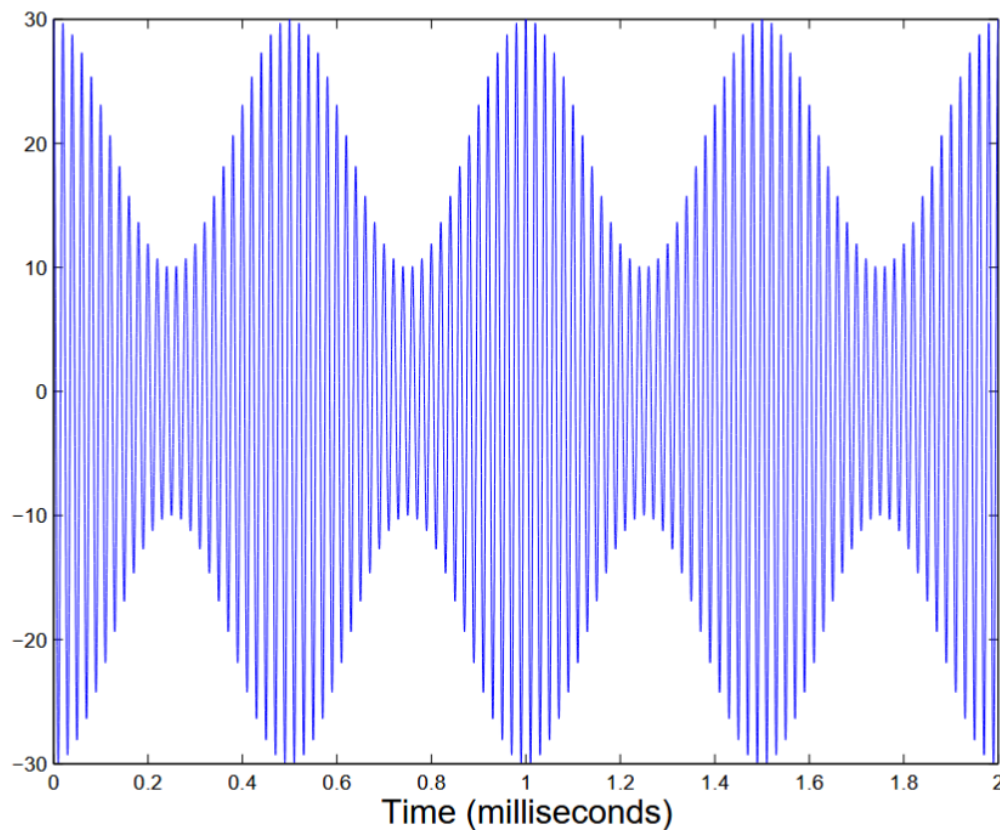


Figure 5: Amplitude Modulated signal

13. Figure 5 shows a signal obtained after amplitude modulation by a sinusoidal message. The carrier frequency is difficult to determine from the figure, and is not needed for answering the questions below.
 - (a) Find the modulation index.
 - (b) Find the signal power.
 - (c) Find the bandwidth of the AM signal.

14. The message

$$m(t) = 2 \sin 2000\pi t - 3 \cos 4000\pi t$$

is used in an AM system with a modulation index of 70% and carrier frequency of 580 kHz. What is the power efficiency? If the net transmitted power is 10 watts, find the magnitude spectrum of the transmitted signal.

15. Consider a square-law detector, using a non-linear device whose transfer function is

$$v_2(t) = a_1 v_1(t) + a_2 v_1^2(t)$$

where a_1 and a_2 are constants and $v_1(t)$ is the input and $v_2(t)$ is the output. Suppose you input an AM signal to the above device, i.e.,

$$v_1(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t).$$

1. Evaluate the output $v_2(t)$.
 2. Find the condition for which the message signal $m(t)$ may be recovered from $v_2(t)$.
16. Let $p(t) = I_{[-\frac{1}{2}, \frac{1}{2}]}(t)$ denote a rectangular pulse of unit duration. Construct the signal

$$m(t) = \sum_{n=-\infty}^{\infty} (-1)^n p(t - n).$$

The signal $m(t)$ is input to an FM modulator, whose output is given by

$$u(t) = 20 \cos(2\pi f_c t + \phi(t))$$

where

$$\phi(t) = 20\pi \int_{-\infty}^t m(\tau) d\tau + a$$

and a is chosen such that $\phi(0) = 0$.

- (a) Carefully sketch both $m(t)$ and $\phi(t)$ as a function of time.
- (b) Approximating the bandwidth of $m(t)$ as $W \approx 2$, estimate the bandwidth of $u(t)$ using Carson's formula.
- (c) Suppose that a very narrow ideal BPF (with bandwidth less than 0.1) is placed at $f_c + \alpha$. For which (if any) of the following choices of α will you definitely get zero power at the output of the BPF (Note: you may or may not get non-zero power for certain α , but the question asks for the α at which you'll certainly get zero power)
 - (i) $\alpha = 0.5$
 - (ii) $\alpha = 0.75$
 - (iii) $\alpha = 1$

17. A phase modulated signal

$$s(t) = A_c \cos(2\pi f_c t + k_p m(t))$$

is demodulated using an ideal differentiator followed by an envelope detector.

1. Show that the output of the differentiator contains a term proportional to the message signal.
2. Identify the high-frequency terms present in the output.
3. Explain how the original message signal is recovered.